

Causality

EC 421, Set 10

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Prologue

Schedule

Last Time

Autocorrelation and nonstationarity

Today

Causality

Causality

Intro

Most tasks in econometrics boil down to one of two goals:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_k x_k + u$$

1. Often called *causal identification*.

For the rest of the term, we will focus on **causally estimating** β_j .

Causality

The challenges

As you saw in the data-analysis exercise, determining and estimating the true model can be pretty difficult—both **practically** and **econometrically**.

Causality

It's complicated

Occasionally, **causal** relationships are simply/easily understood, e.g.,

Causality

Correlation $\neq$ Causation

You've likely heard the saying

| Correlation is not causation.

The saying is just pointing out that there are violations of exogeneity.

Let's work through a few examples.

Causation

Example: The causal effect of fertilizer

Suppose we want to know the causal effect of fertilizer on corn yield.¹

1. Many of the early statistical and econometric studies involved agricultural field trials.

Causation

Example: The causal effect of fertilizer

Randomized experiments help us maintain *all else equal* (exogeneity).

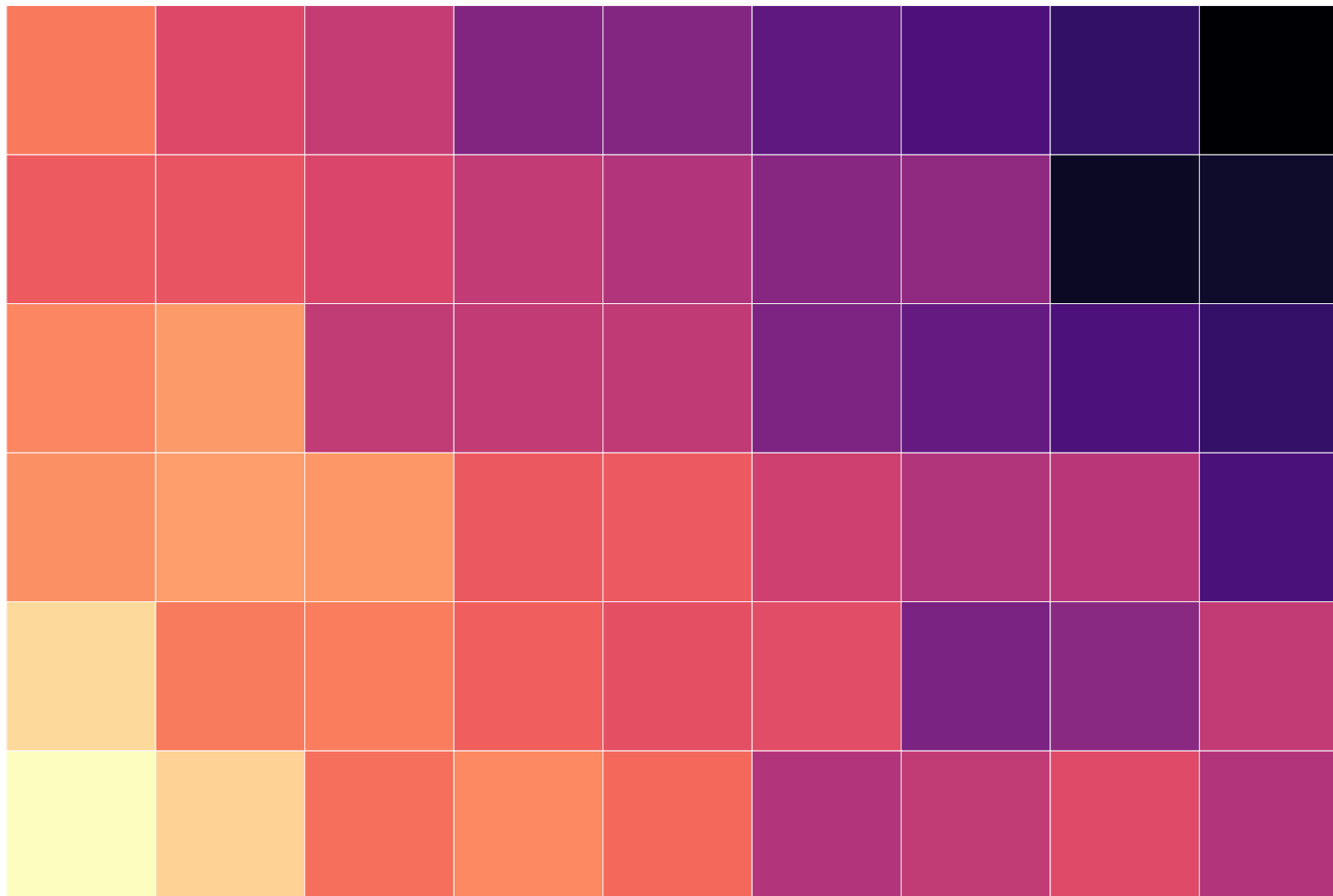
We often call these experiments **randomized control trials** (RCTs).¹

1. Econometrics (and statistics) borrows this language from biostatistics and pharmaceutical trials.

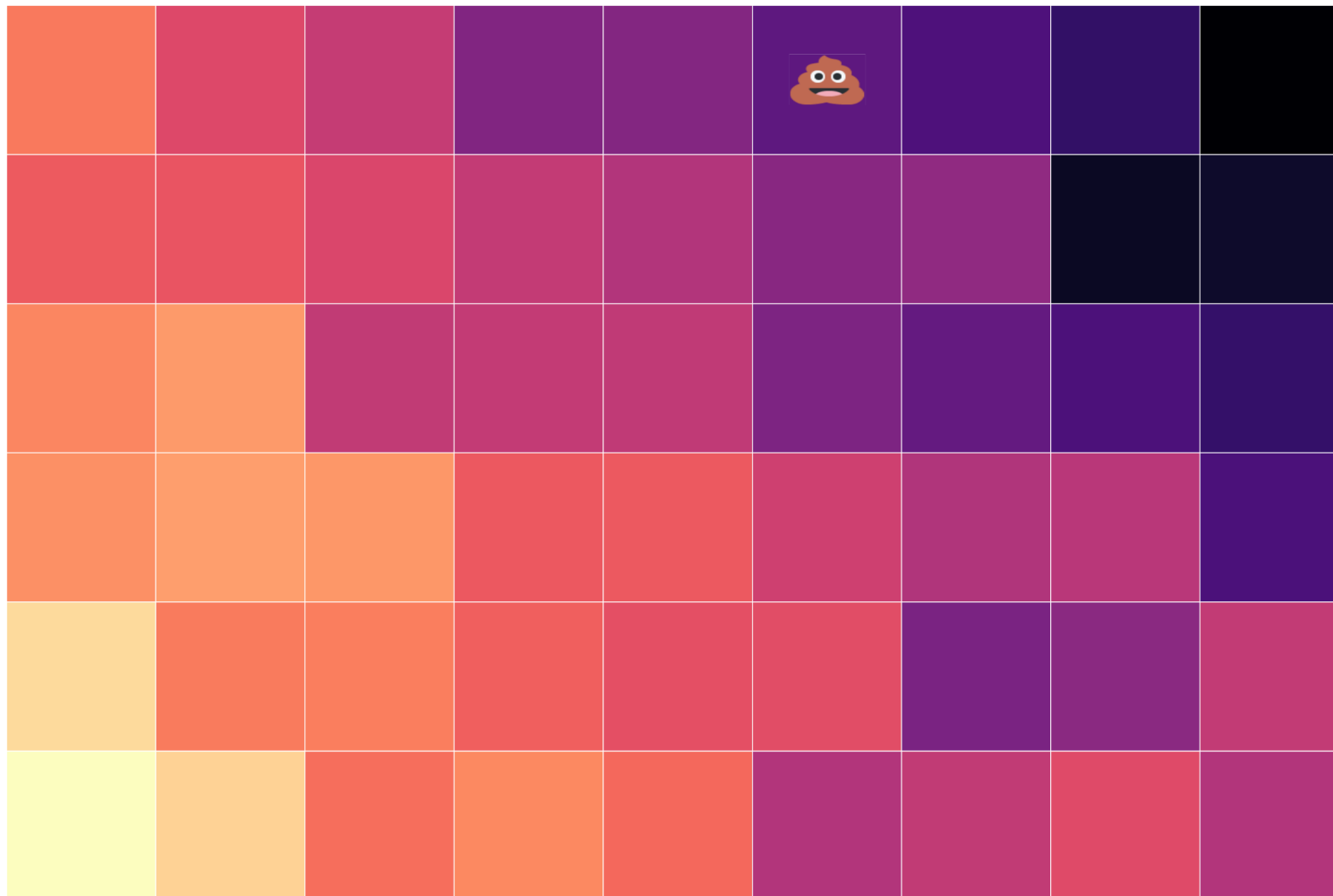
54 equal-sized plots

01	02	03	04	05	06	07	08	09
10	11	12	13	14	15	16	17	18
19	20	21	22	23	24	25	26	27
28	29	30	31	32	33	34	35	36
37	38	39	40	41	42	43	44	45
46	47	48	49	50	51	52	53	54

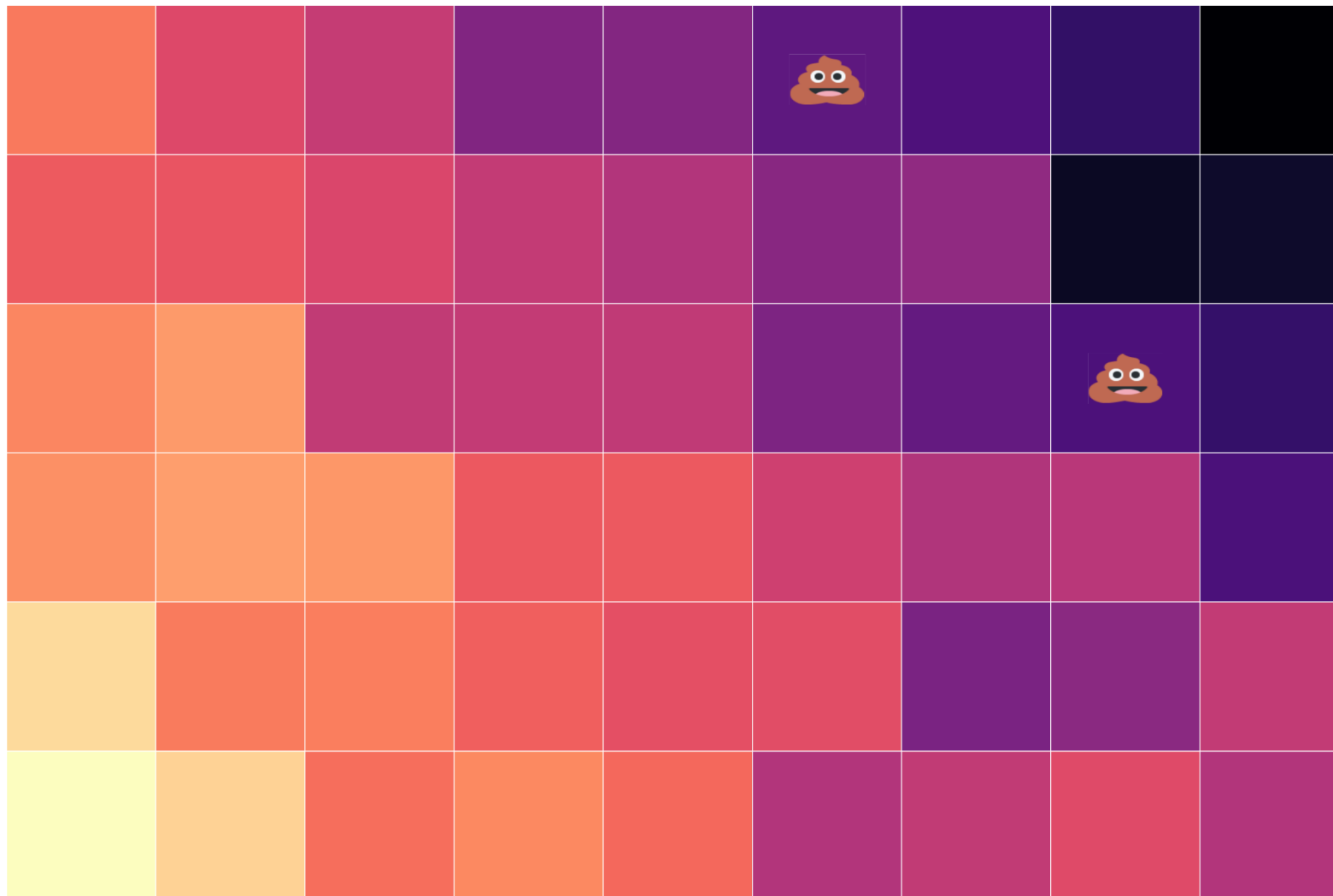
54 equal-sized plots



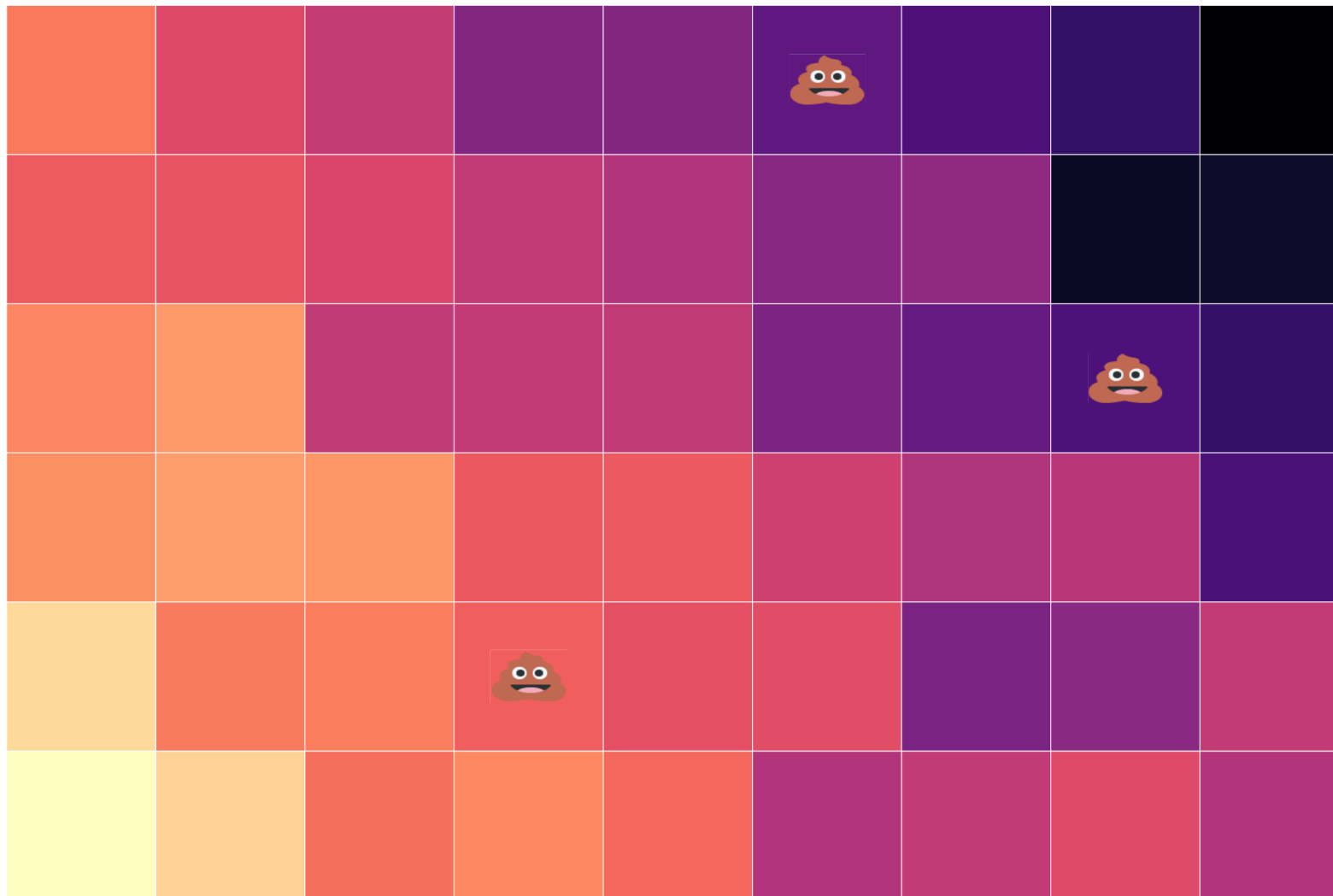
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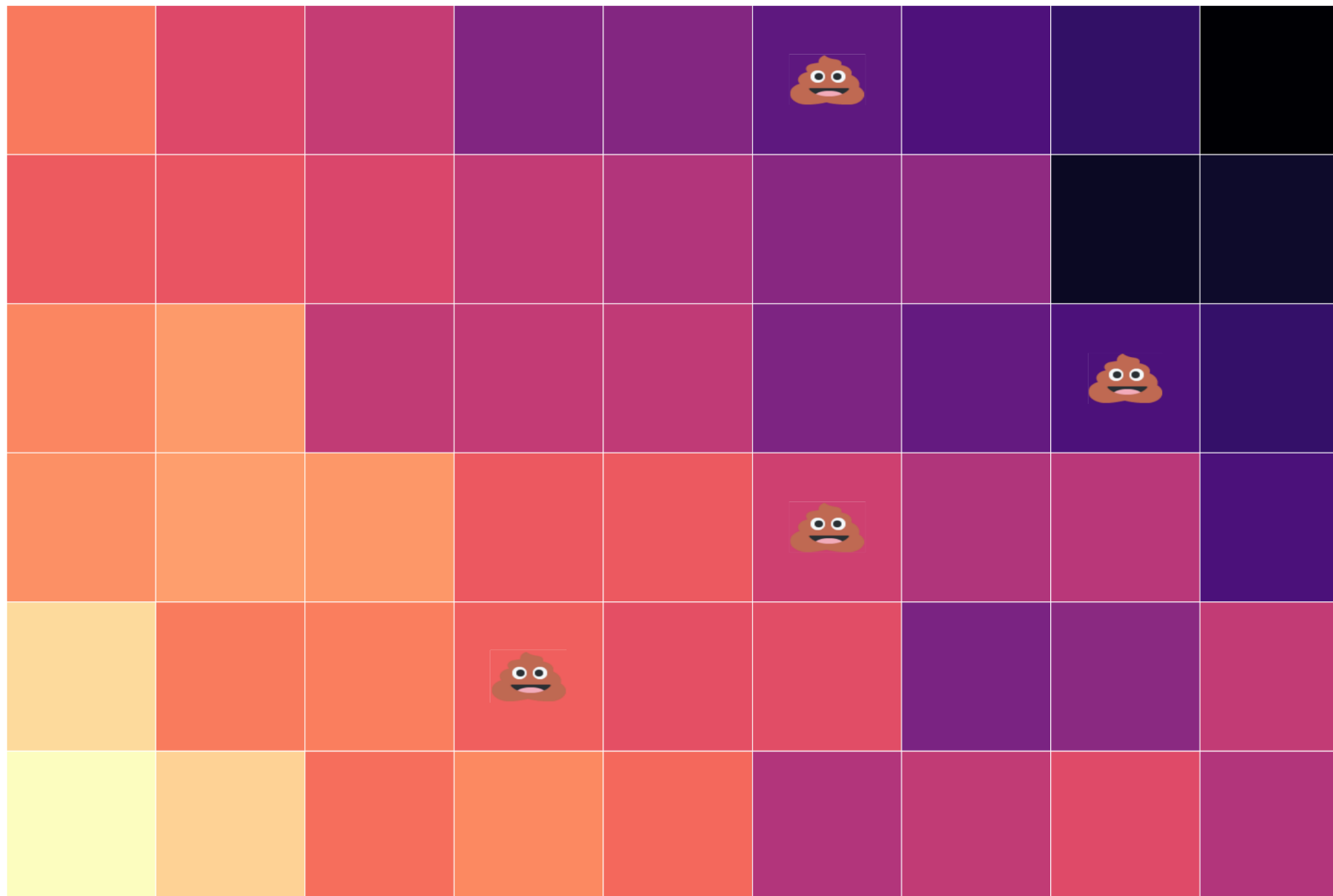
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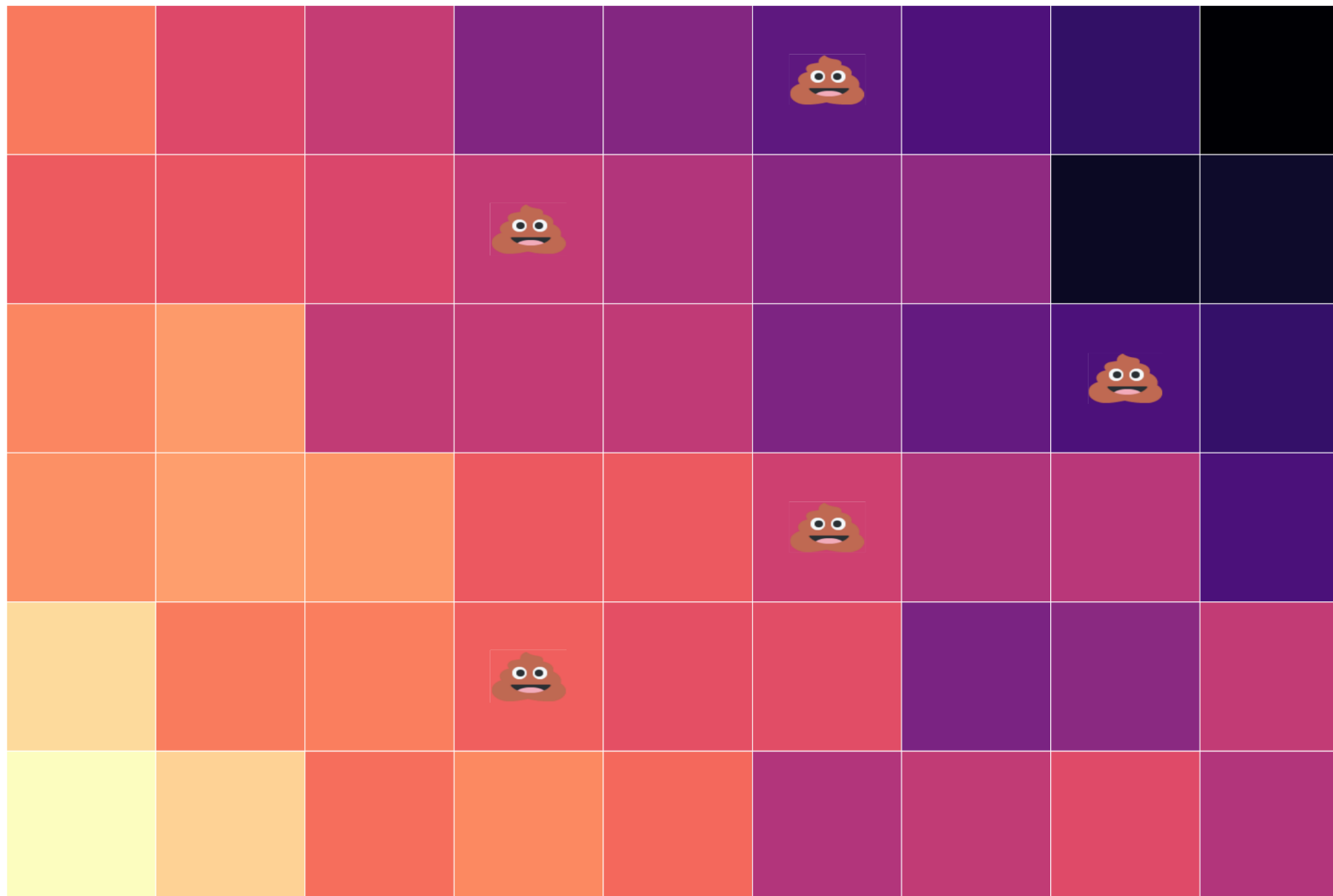
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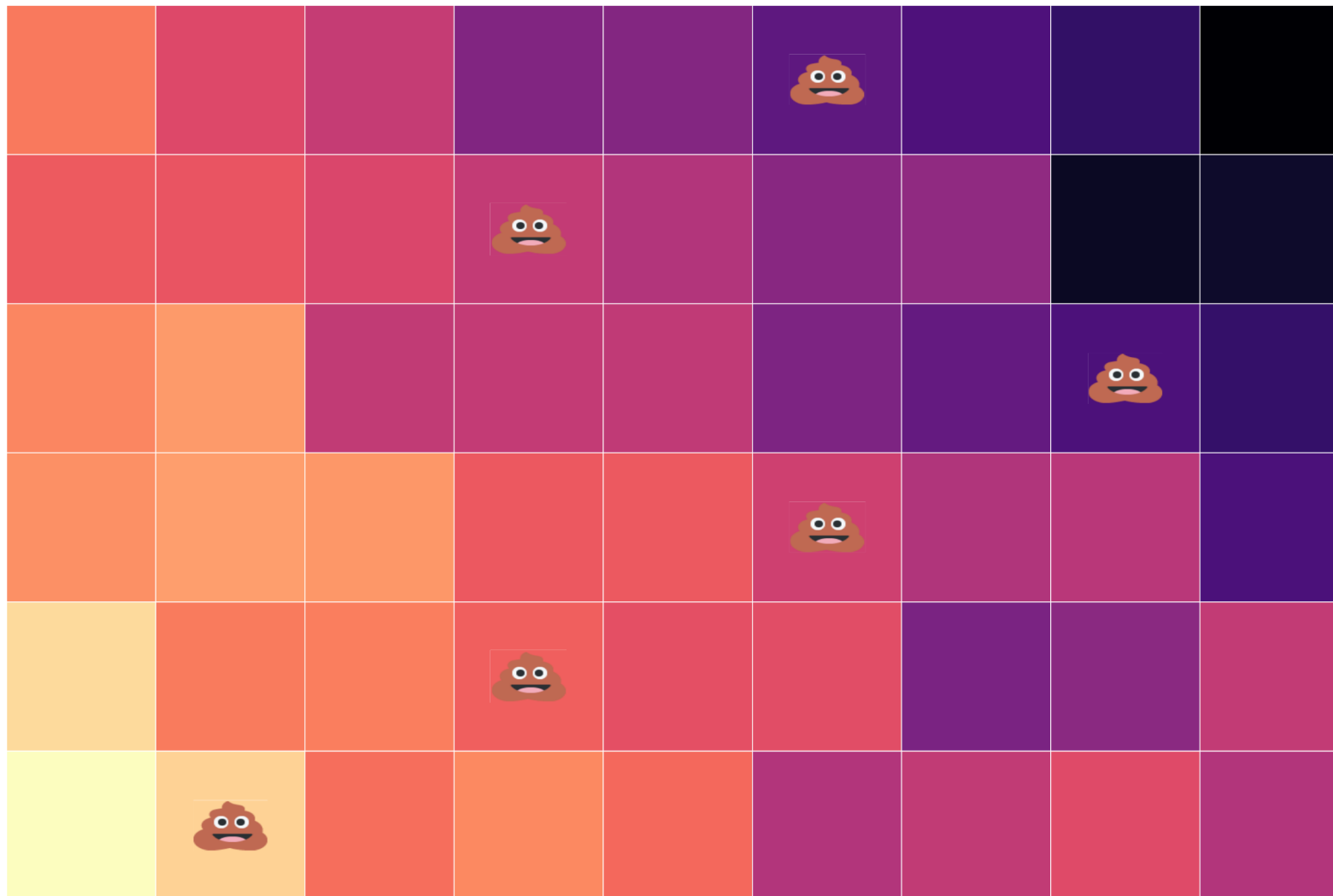
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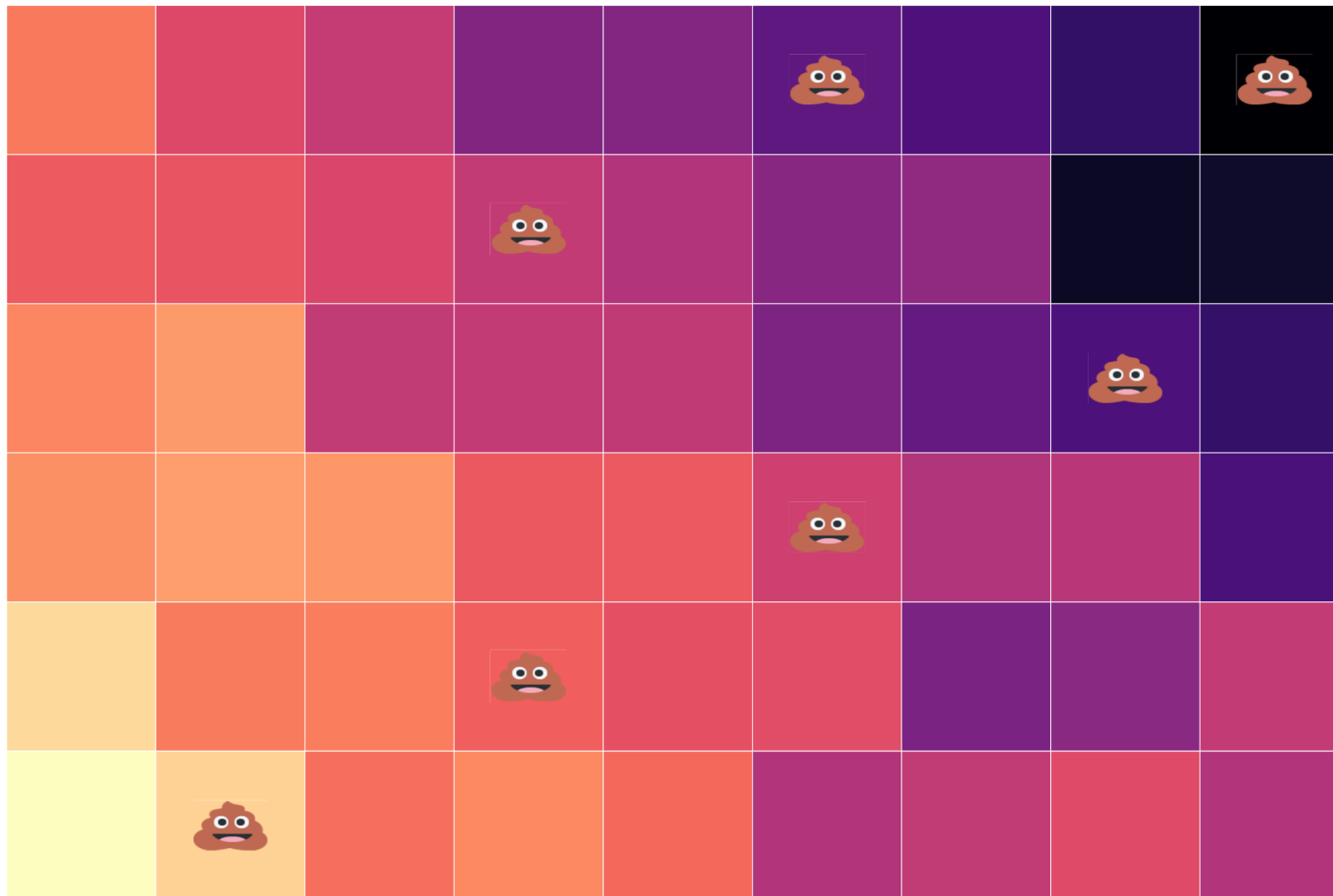
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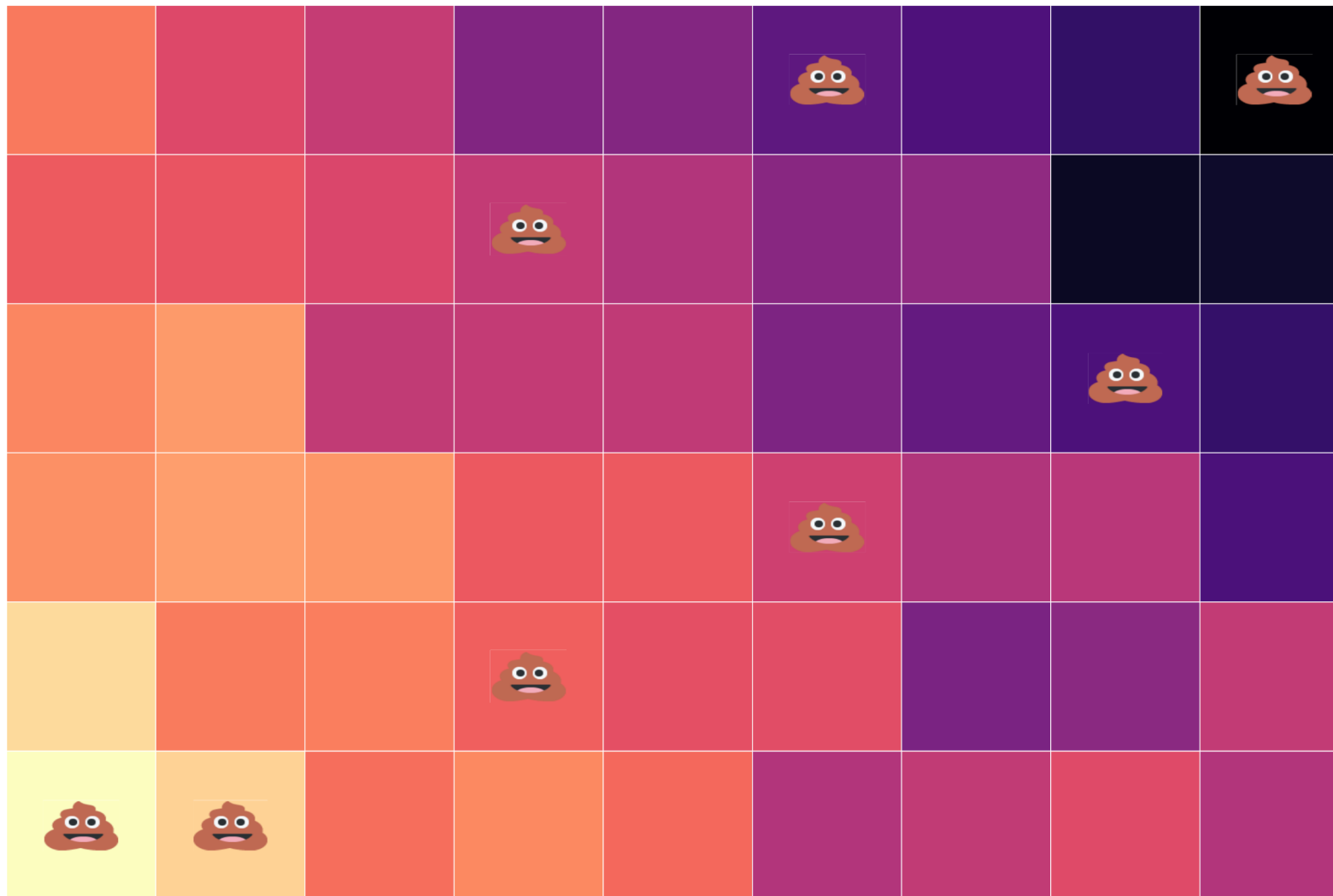
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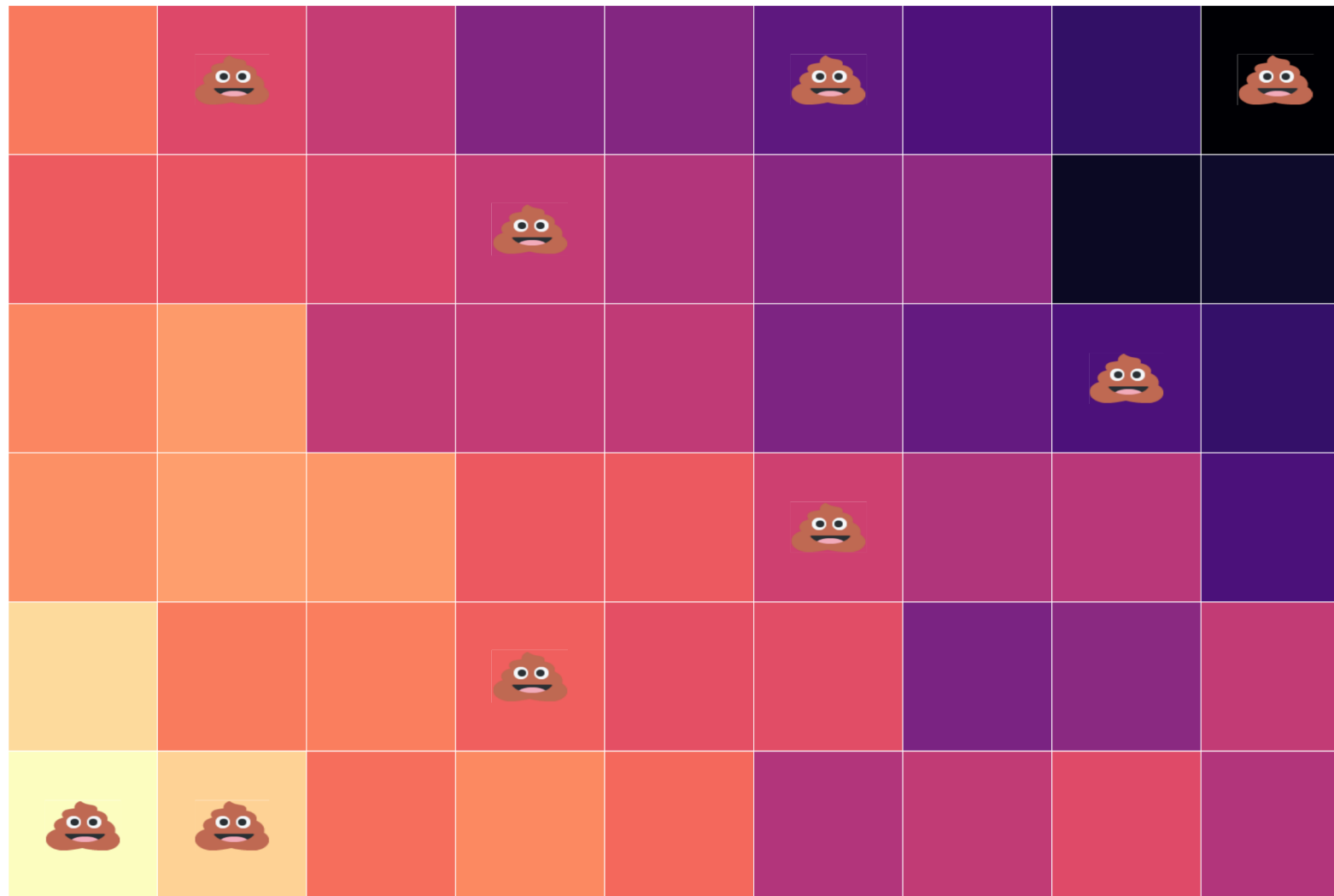
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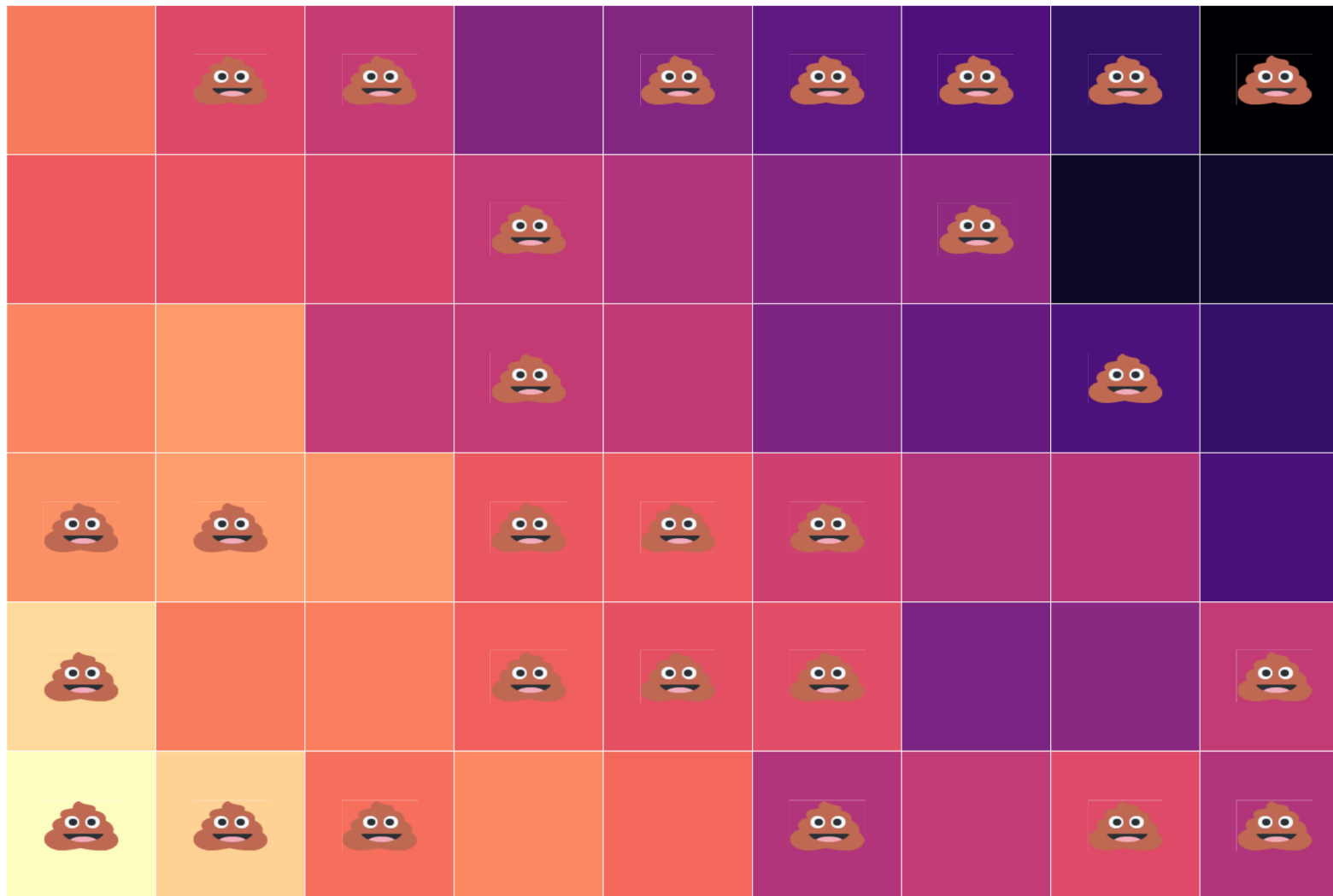
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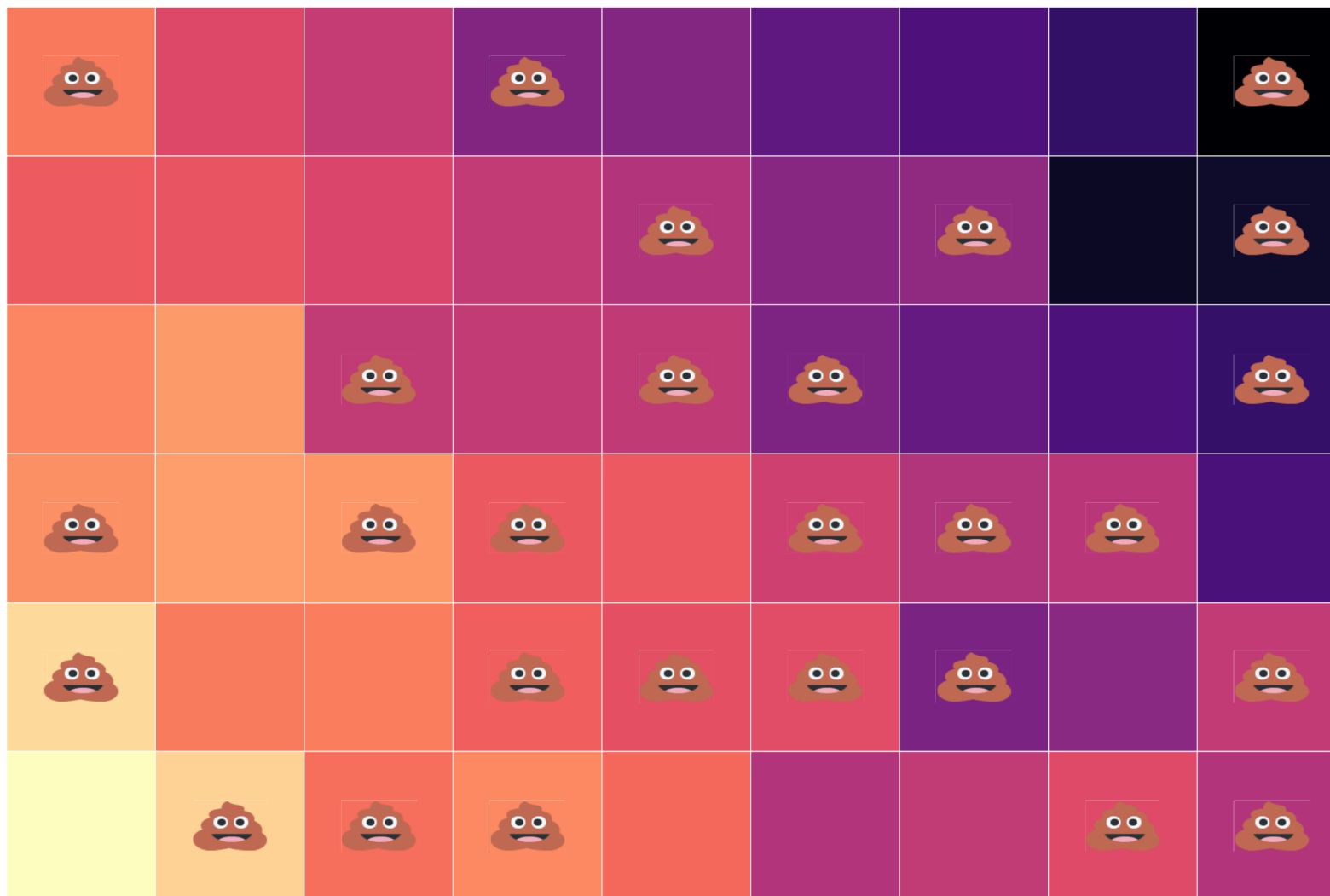
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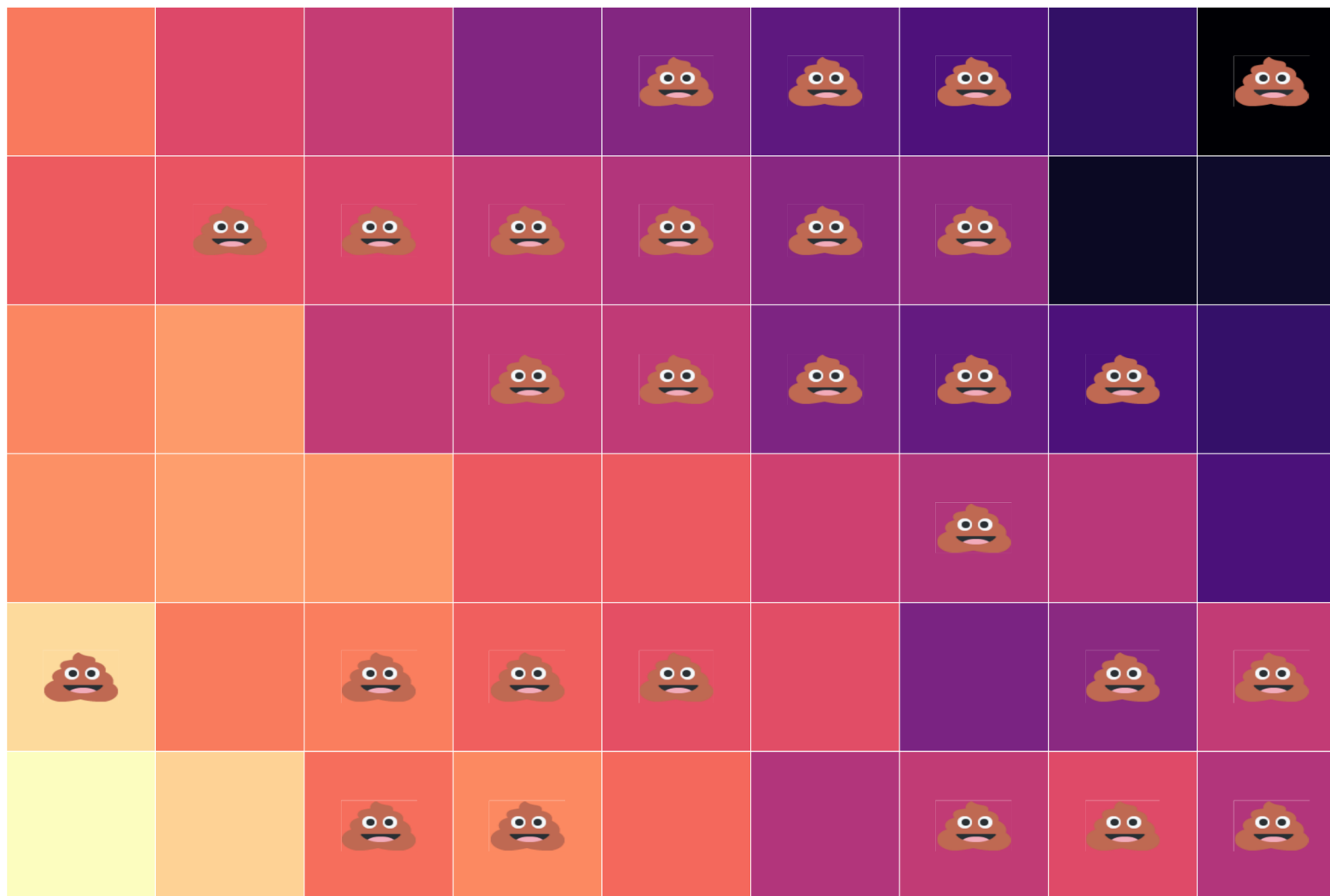
54 equal-sized plots



54 equal-sized plots



54 equal-sized plots



Causation

Example: The causal effect of fertilizer

We can estimate the **causal effect** of fertilizer on crop yield by comparing the average yield in the treatment group (💩) with the control group (no 💩).

$$\overline{\text{Yield}}_{\text{Treatment}} - \overline{\text{Yield}}_{\text{Control}}$$

Causality

Example: Returns to education

Labor economists, policy makers, parents, and students are all interested in the (monetary) *return to education*.

Causality

Example: Returns to education

Q: Could we simply regress earnings on education?

Causality

Example: Returns to education

Q: So how can we estimate the returns to education?

Causality

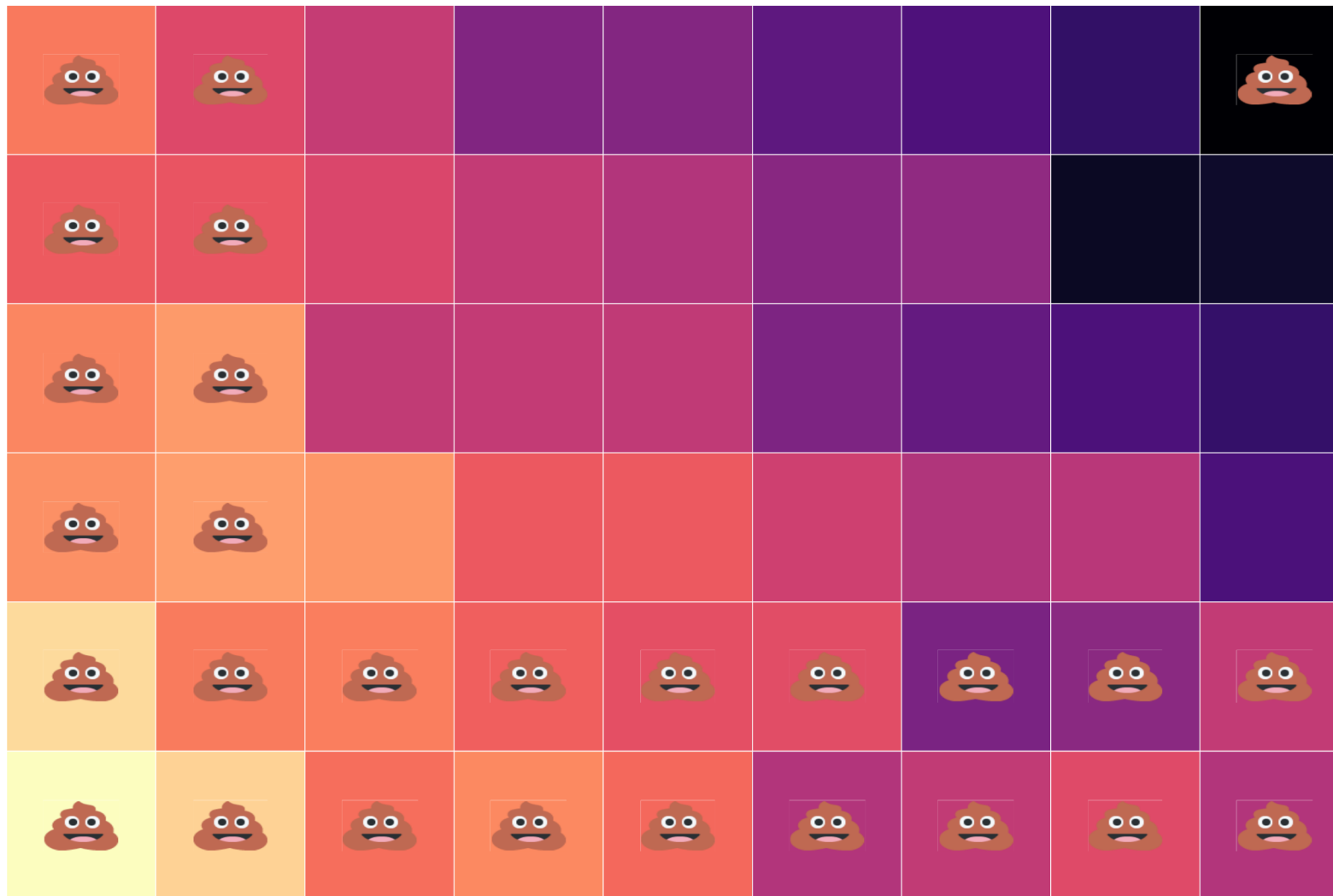
Real-world experiments

Both examples consider **real experiments** that isolate causal effects.

Characteristics

- Feasible—we can actually (potentially) run the experiment.
- Compare individuals randomized into treatment against individuals randomized into control.
- Require “good” randomization to get *all else equal* (exogeneity).

Unfortunate randomization



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The ideal experiment

The absolute, best-case scenario would be somewhat different.

Rather than comparing units randomized as **treatment** vs. **control**, we would compare treatment and control **for the same, exact unit**.

1. Without (1) God-like abilities and multiple universes or (2) a time machine.

Causality

The ideal experiment

The *ideal* data for 10 people

```
#>      i trt  y1i  y0i
#> 1    1   1 5.01 2.56
#> 2    2   1 8.85 2.53
#> 3    3   1 6.31 2.67
#> 4    4   1 5.97 2.79
#> 5    5   1 7.61 4.34
#> 6    6   0 7.63 4.15
#> 7    7   0 4.75 0.56
#> 8    8   0 5.77 3.52
#> 9    9   0 7.47 4.49
#> 10  10   0 7.79 1.40
```

Causality

The ideal experiment

The *ideal* data for 10 people

```
#>      i trt  y1i  y0i effect_i
#> 1    1   1  5.01  2.56      2.45
#> 2    2   1  8.85  2.53      6.32
#> 3    3   1  6.31  2.67      3.64
#> 4    4   1  5.97  2.79      3.18
#> 5    5   1  7.61  4.34      3.27
#> 6    6   0  7.63  4.15      3.48
#> 7    7   0  4.75  0.56      4.19
#> 8    8   0  5.77  3.52      2.25
#> 9    9   0  7.47  4.49      2.98
#> 10  10   0  7.79  1.40      6.39
```

Calculate the causal effect of trt.

$$\tau_i = y_{1,i} - y_{0,i}$$

for each individual i .

Causality

The ideal experiment

The *ideal* data for 10 people

```
#>      i trt  y1i  y0i effect_i
#> 1    1   1  5.01  2.56      2.45
#> 2    2   1  8.85  2.53      6.32
#> 3    3   1  6.31  2.67      3.64
#> 4    4   1  5.97  2.79      3.18
#> 5    5   1  7.61  4.34      3.27
#> 6    6   0  7.63  4.15      3.48
#> 7    7   0  4.75  0.56      4.19
#> 8    8   0  5.77  3.52      2.25
#> 9    9   0  7.47  4.49      2.98
#> 10  10   0  7.79  1.40      6.39
```

Calculate the causal effect of trt.

$$\tau_i = y_{1,i} - y_{0,i}$$

for each individual i .

The mean of τ_i is the

average treatment effect (ATE).

Thus, $\bar{\tau} = 3.82$

Causality

The ideal experiment

This model highlights the fundamental problem of causal inference.

$$\tau_i = y_{1,i} - y_{0,i}$$

Causality

The ideal experiment

So a dataset that we actually observe for 6 people will look something like

```
#>      i trt  y1i  y0i
#> 1    1   1 5.01   NA
#> 2    2   1 8.85   NA
#> 3    3   1 6.31   NA
#> 4    4   1 5.97   NA
#> 5    5   1 7.61   NA
#> 6    6   0  NA 4.15
#> 7    7   0  NA 0.56
#> 8    8   0  NA 3.52
#> 9    9   0  NA 4.49
#> 10  10   0  NA 1.40
```

Causality

Causally estimating the treatment effect

Notation:

- $D_i = 1$ if individual i is treated.
- $D_i = 0$ if individual i is not treated (*control* group).

Causality

Causally estimating the treatment effect

Q: How can we estimate $\bar{\tau}$ using only $(y_{1,i} | D_i = 1)$ and $(y_{0,i} | D_i = 0)$?

Causality

Causally estimating the treatment effect

Assumption:

This assumption says that the treatment effect is equal (constant) across all individuals i .

Q_{3.0}: Is $Avg(y_i \mid D_i = 1) - Avg(y_i \mid D_i = 0)$ a *good* estimator for τ ?

Selection into treatment

So our proposed group-difference estimator give us the sum of

1. τ , the **causal, average treatment effect** that we want
2. **Selection bias**: How much trt. and control groups differ (on average).

Randomization shuts down selection bias (on average).

Without randomization, we need to be careful.

Next time: Solving selection bias.

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